

Hellenic Complex Systems Laboratory

Enteral Nutritional Design

Technical Report X

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Enteral Nutritional Design

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Short Description of the Demonstration

Enteral feeding refers to the delivery of nutrition directly into the gastrointestinal tract, thereby providing part or all of a patient's energy and nutrient requirements. To design an enteral nutrition formula, the program calculates the quantities of up to six selected food items: one meat or fish item, one cereal, one dairy food, one vegetable, one fruit, and one oil. The combination is required to comply with user-defined specifications: total required energy, fraction of energy from protein, fraction of energy from carbohydrate, and maximum total water content. The specifications are set using sliders, and the foods are selected using their respective menus. Each menu includes a "None" option.

Search Terms

nutrition, nutrition therapy, enteral nutrition, nutrition design, nutrition formula, composition of foods, critical care, intensive care, medicine

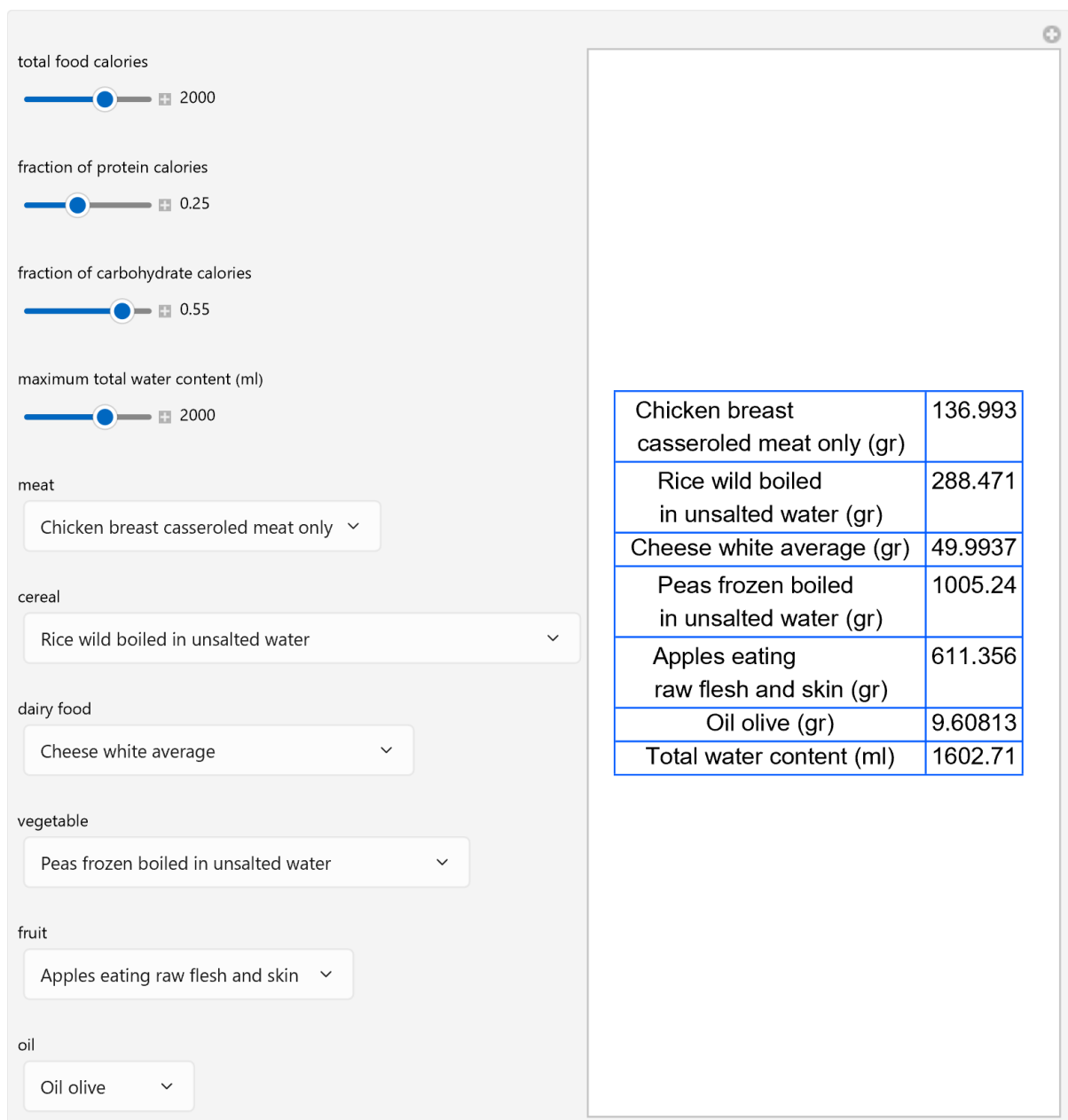


Figure 1: Calculated enteral nutrition formula showing the selected food items, their required quantities, and total water content, with the settings shown on the left.

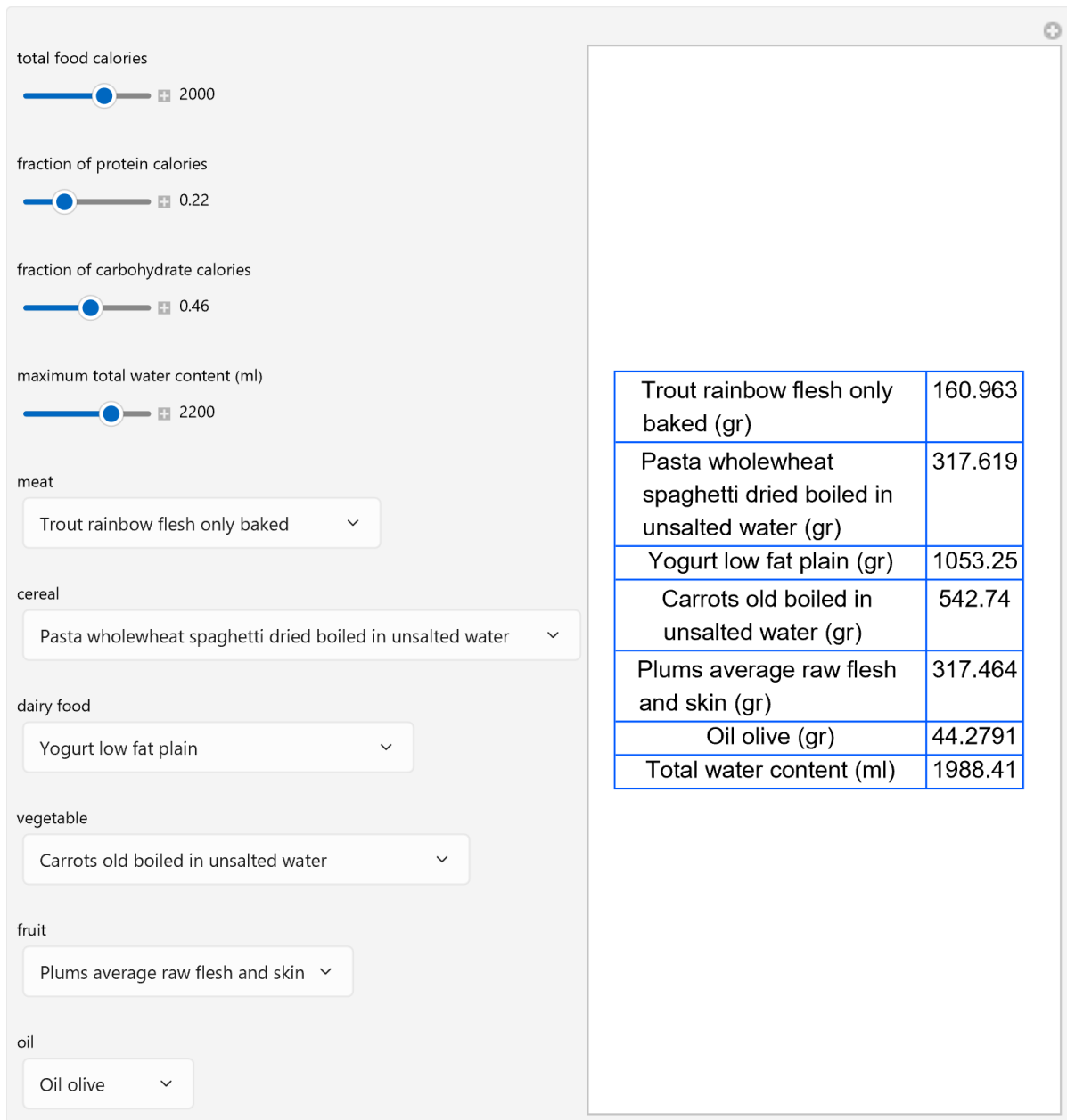


Figure 2: Calculated enteral nutrition formula showing the selected food items, their required quantities, and total water content, with the settings shown on the left.

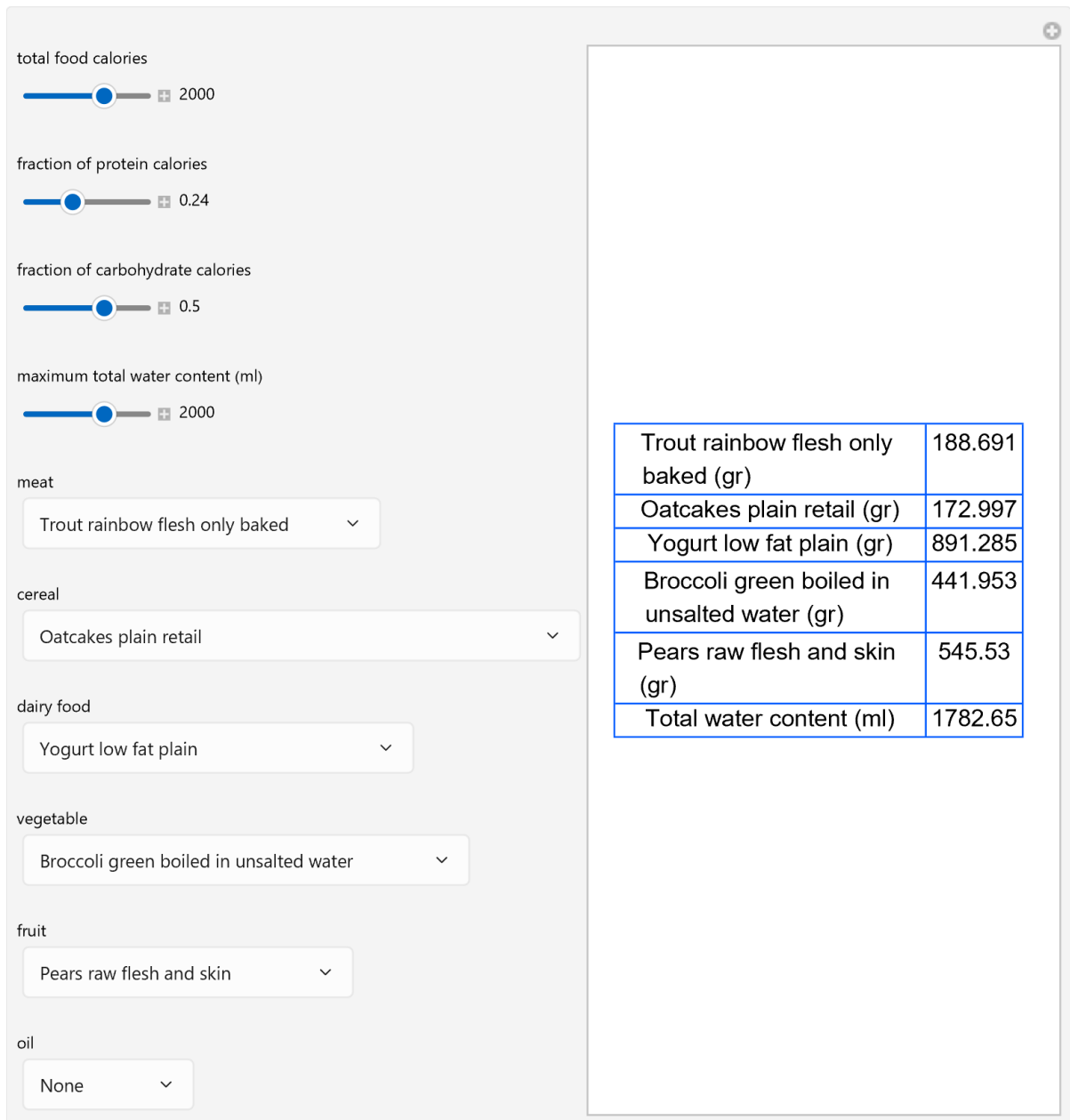


Figure 3: Calculated enteral nutrition formula showing the selected food items, their required quantities, and total water content, with the settings shown on the left.

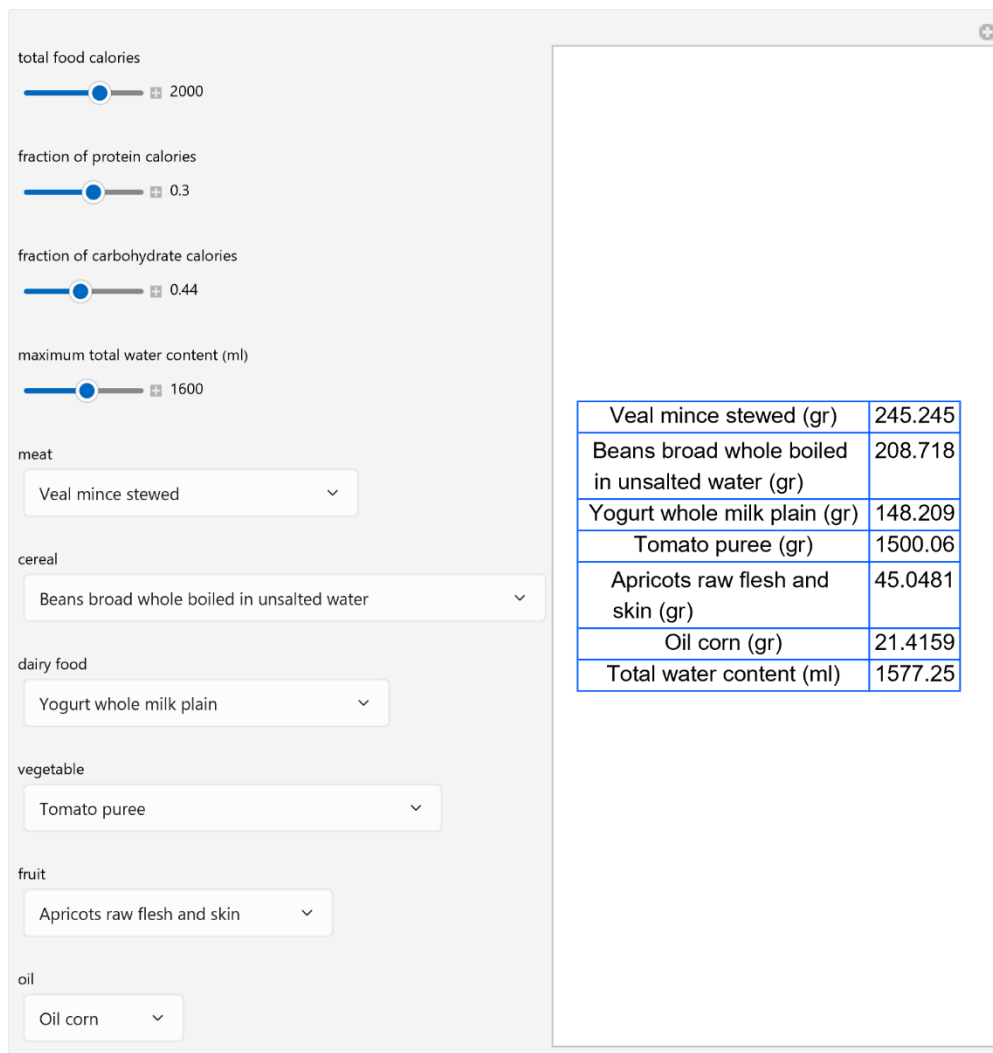


Figure 4: Calculated enteral nutrition formula showing the selected food items, their required quantities, and total water content, with the settings shown on the left.

Details

This Demonstration shows a method for the design of enteral nutrition [1]. It includes the nutrient composition data of all the foods included in the 2015 version of the Composition of Foods Integrated Dataset (CoFID), published by Public Health England (PHE) [2].

Accordingly, the menus of the Manipulate function can be redefined to include any of the foods of the dataset.

References

- [1] R. Bankhead, J. Boullata, S. Brantley, M. Corkins, P. Guenter, J. Krenitsky, B. Lyman, N. A. Metheny, C. Mueller, S. Robbins, J. Wessel and the A.S.P.E.N. Board of Directors. A.S.P.E.N. Enteral Nutrition Practice Recommendations. *Journal of Parenteral and Enteral Nutrition*, **33**(2), 2009 pp. 122–167.
doi:10.1177/0148607108330314.
- [2] P. M. Finglas, M. A. Roe, H. M. Pinchen, R. Berry, S. M. Church, S. K. Dodhia, M. Farron-Wilson and G. Swan, McCance and Widdowson's The Composition of Foods. 7th summary ed., Cambridge, UK: The Royal Society of Chemistry, 2015.

Demonstration

DOI

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Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/EnteralNutritionalDesign.nb>

First published in 2017 as part of the [Wolfram Demonstrations Project](https://demonstrations.wolfram.com/):

<https://demonstrations.wolfram.com/EnteralNutritionalDesign/>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

Citation

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